21. Kiefer-Wolfowitz

March 4, 2026 9:50 AM

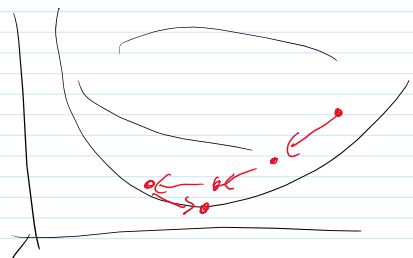
The week's Topic: Stochastic Approximation

- Minimize a (noisy) loss function

  $L(\theta)$: Noise-free loss

  $Q(\theta, v)$: Noisy loss

  $y(\theta)$: Noisy gradient



- Methods $\Rightarrow$ compute the gradient

  $\Rightarrow$ walk down hill, i.e. follow the gradient to look for a smaller loss

- Robbins-Monroe: $\theta_{k+1} = \theta_k - a_k\, y(\theta_k)$

  new, old, "gain", Gradient

want $a_k \to 0$, but not too fast

- $\sum_{k=1}^{\infty} a_k = \infty$, $\quad$ • $\sum_{k=1}^{\infty} a_k^2 < \infty$

Motivating (simple) example:

$$\bar{X}_{n+1} = \bar{X}_n - \frac{1}{n+1}\underbrace{(\bar{X}_n - X_{n+1})}_{[(\bar{X}_n - \mu) + (\mu - X_{n+1})]} = \theta_n - \frac{1}{n+1}\overbrace{(\theta_n - X_{n+1})}^{y(\theta_n)}$$

$(\bar{X}_n - \mu)$: gradient $\quad + \quad$ $(\mu - X_{n+1})$: Mean-zero Noise

RM requires a gradient computation

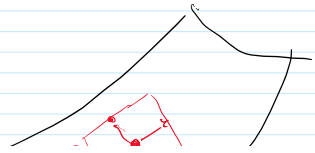
- What if we can't compute the gradient of $Q(\theta, v)$?
- Solution: Approximate the gradient

$$\theta_{k+1} = \theta_k - a_k\, \hat{y}(\theta_k)$$

- How should we approx the gradient?

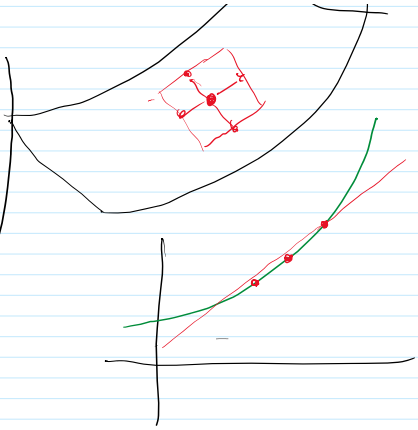
<u>Kiefer-Wolfowitz</u> (1950s)

$$\frac{1}{\ldots}\,(Q(\theta_k + c_k e_\ldots) - Q(\theta_k - c_k e_\ldots))$$



Kiefer-Wolfowitz (1950s)

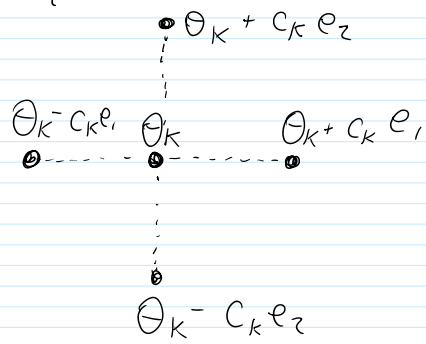
$$\hat{y}_k(\theta_k) = \frac{1}{2c_k}\begin{pmatrix} Q(\theta_k + c_k e_1) - Q(\theta_k - c_k e_1) \\ \vdots \\ Q(\theta_k + c_k e_d) - Q(\theta_k - c_k e_d) \end{pmatrix}$$



$$e_i = (\underbrace{0, \dots, 0}_{0}, \underset{\substack{\uparrow \\ i\text{th} \\ \text{entry}}}{1}, \underbrace{0, \dots, 0}_{0})$$

c_k is another "gain" or "step-size" sequence

→ if Q is "nice" enough, then picking points around θ_k and approximating the gradient should be close to the true gradient.



- require $a_k > 0$, $c_k > 0$, $\forall k$
- $a_k \to 0$, $c_k \to 0$
- $\sum_{k=1}^{\infty} a_k = \infty$, $\sum_{k=1}^{\infty} a_k c_k < \infty$, $\sum_{k=1}^{\infty} a_k^2 / c_k^2 < \infty$

Why? to make the theory on convergence of KW work out.

Roughly, $a_k \to 0$ slow enough
and $c_k \to 0$ slower than a_k

Example: $a_k = 1/k$, $c_k = 1/k^{\alpha}$ for $\alpha \in (0, 1/2)$

One Problem with KW

→ in d-dimensions, we have to evaluate $Q(\cdot)$ at $2d$ points at every iteration!

- Q may be hard (computationally) to compute

→ Alternatives to KW

- Random perturbation

$$\hat{y}(\theta_k) = \frac{[Q(\theta_k + c_k v) - Q(\theta_k - c_k v)]}{2c_k}\begin{pmatrix} v_1 \\ \vdots \\ v_d \end{pmatrix}$$

$$y(\theta_k) = \frac{[Q(\theta_k + c_k v) - Q(\theta_k - c_k v)]}{2c_k} \begin{pmatrix} \vdots \\ v_d \end{pmatrix}$$

where v is some random vector

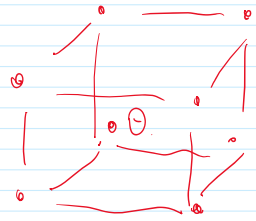
say $v \sim N(0, \sigma^2 I)$

Pro: we only need 2 evaluations of Q at each step.

Con: v is a random direction which may be "good" or "bad"

- Simultaneous Perturbation

same idea but $v = \begin{pmatrix} \pm 1 \\ \vdots \\ \pm 1 \end{pmatrix}$



pick a ~~point~~ vertex of the Hyper cube about θ_k